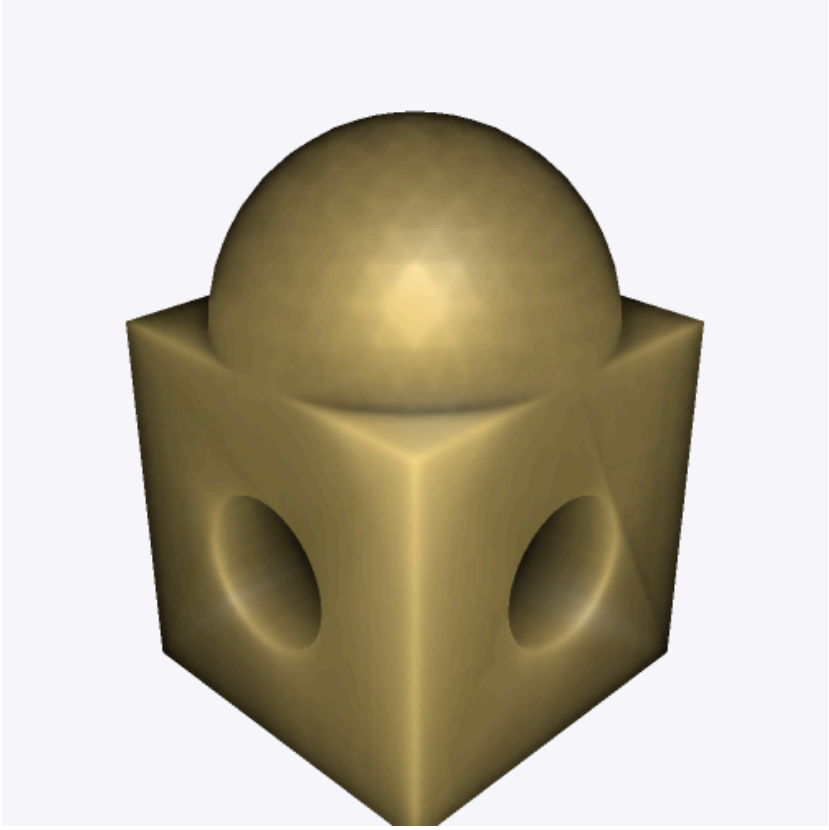
maquette-scad

Parametric CAD in Typst, via OpenSCAD + Manifold

github.com/bernsteining/maquette · bernsteining.github.io/maquette



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Introduction

maquette-scad extends `maquette` with procedural geometry: takes an OpenSCAD-flavored expression (or a `.scad` source string), compiles it to a watertight mesh via the **Manifold** CSG kernel, and returns PLY bytes that plug straight into maquette's `render-ply` — all inside Typst's wasm sandbox. No external OpenSCAD install, no shell-out, no intermediate `.stl` on disk.

Two entry points:

- **Typst DSL** — write geometry as Typst expressions using `cube`, `sphere`, `translate`, `difference` etc. Terse, composable, uses Typst's own `for` / `range` / `calc` for procedural placement.
- **`.scad` ingestion** — hand in the text of an existing `.scad` file (`read("part.scad")`). Full OpenSCAD language surface: `function`, `module`, `for`, list comprehensions, `include` / `use`, `$fn` / `$fa` / `$fs`.

The CSG kernel guarantees watertight boolean output — none of the "20% of faces open on any non-trivial cut" issue you get from BSP-based kernels.

This document only covers the **geometry** layer — primitives, booleans, transforms, `.scad` ingestion, language coverage. Once you have PLY bytes, everything downstream (camera, lighting, materials, shadows, tone mapping, …) is maquette's job — see maquette's documentation for those knobs. See `crates/maquette-scad/FIDELITY.md` for the exhaustive OpenSCAD language coverage tracker.

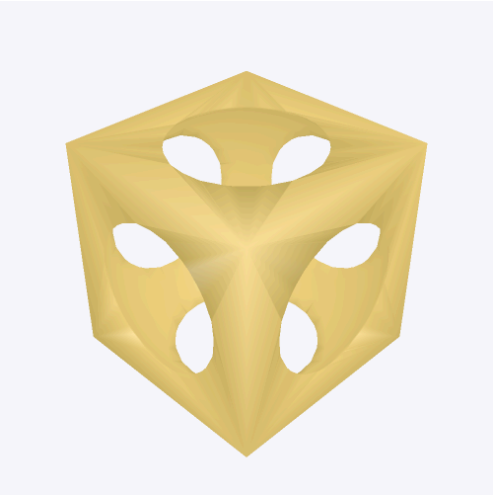
Where to find sample `.scad` files

The DSL examples below run inline — no external files needed. To exercise the `.scad` ingestion path (`compile-scad(read(...))`), grab source from:

- openscad/openscad `examples/` — the official example set that ships with the OpenSCAD editor.
- BOSL2 — comprehensive OpenSCAD utility library with hundreds of documented example fragments.
- Thingiverse (openscad tag) — large community archive; many things ship the `.scad` source alongside the `.stl`.

Quickstart

```
#let part = scadypst(
  difference(
    cube(20, center: true),
    sphere(12, fn: 48),
  )
)
#show-part(part)
```



Two things to notice:

- `cube` / `sphere` / `difference` build an expression tree of dicts — nothing runs yet.
- `scadypst(tree)` sends the tree to the wasm compiler, which returns PLY bytes.
- `show-part` in this doc is a thin wrapper around `#render-ply(bytes, ..config)` — you'd inline the render config yourself in real use.

Primitives

3D

```
#show-part(scadypst(cube(20, center: true)))
```

```
#show-part(scadypst(sphere(15, fn: 48)))
```

```
#show-part(scadypst(cylinder(30, r: 10, center: true, fn: 48)))
```

Cone / frustum: pass `r1` and `r2` instead of `r`.

```
#show-part(scadypst(cylinder(30, r1: 15, r2: 5, center: true,
fn: 48)))
```

2D → 3D

`square` / `circle` / `polygon` on their own return a 2D shape. `linear-extrude` and `rotate-extrude` lift them into 3D.

```
#show-part(scadypst(linear-extrude(20, star(5, 15, 6))))
```

`linear-extrude` supports a twist (degrees over the full height) and a taper `scale`:

```
#show-part(scadypst(linear-extrude(30, square(15, center: true),
twist: 90, scale: 0.5)))
```

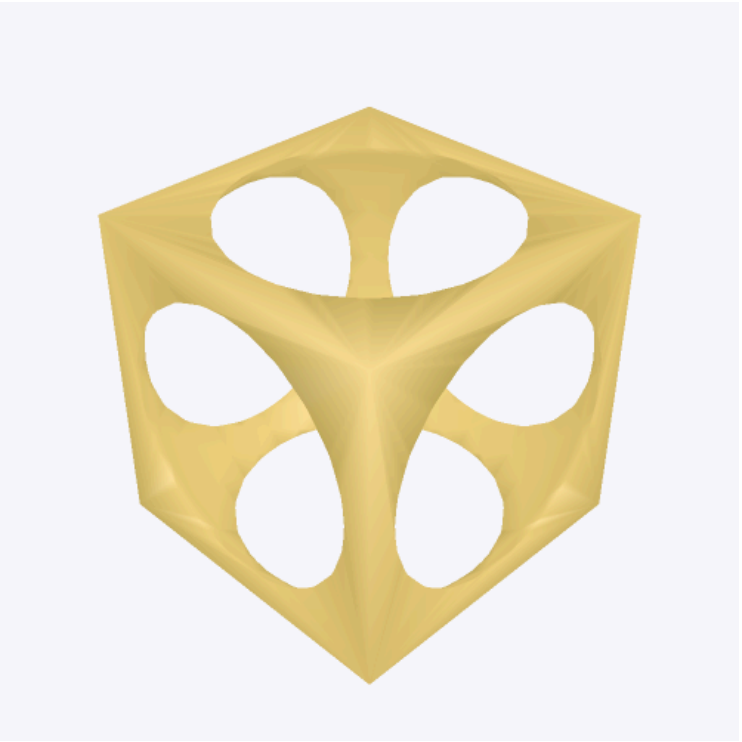
`rotate-extrude` revolves a 2D profile (living in the +x half-plane) around the z-axis. Classic torus:

```
#show-part(scadypst(rotate-extrude(translate((15, 0, 0),
circle(5, fn: 32)), fn: 64)))
```

Boolean operations

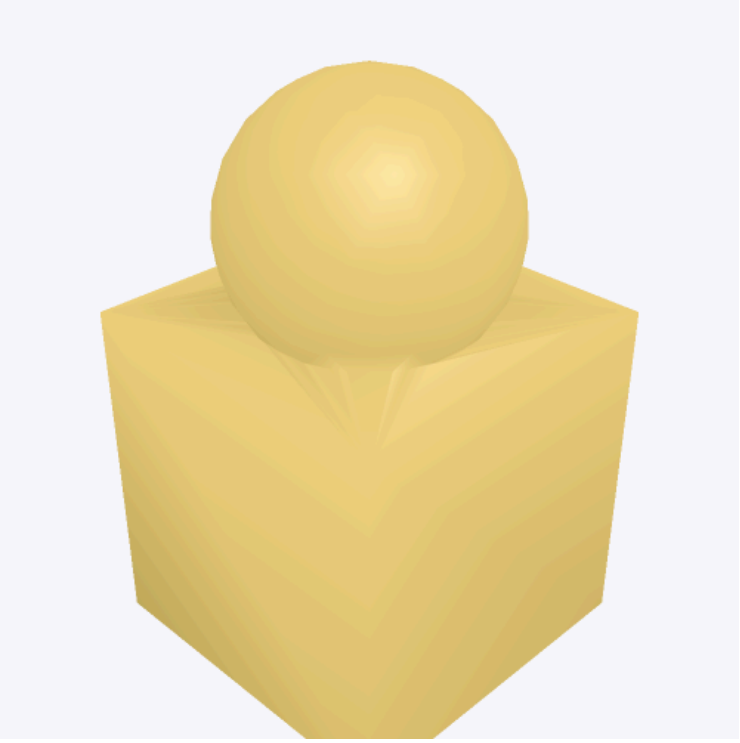
`union`, `difference`, `intersection` — variadic, take any number of children. `difference` is first-minus-the-rest.

```
#show-part(scadypst(
  difference(
    cube(20, center: true),
    sphere(13, fn: 48),
  )
))
```



```
#show-part(scadypst(
  intersection(
    cube(20, center: true),
    sphere(13, fn: 48),
  )
))
```

```
#show-part(scadypst(
  union(
    cube(20, center: true),
    translate((0, 0, 15), sphere(8, fn: 32)),
  )
))
```



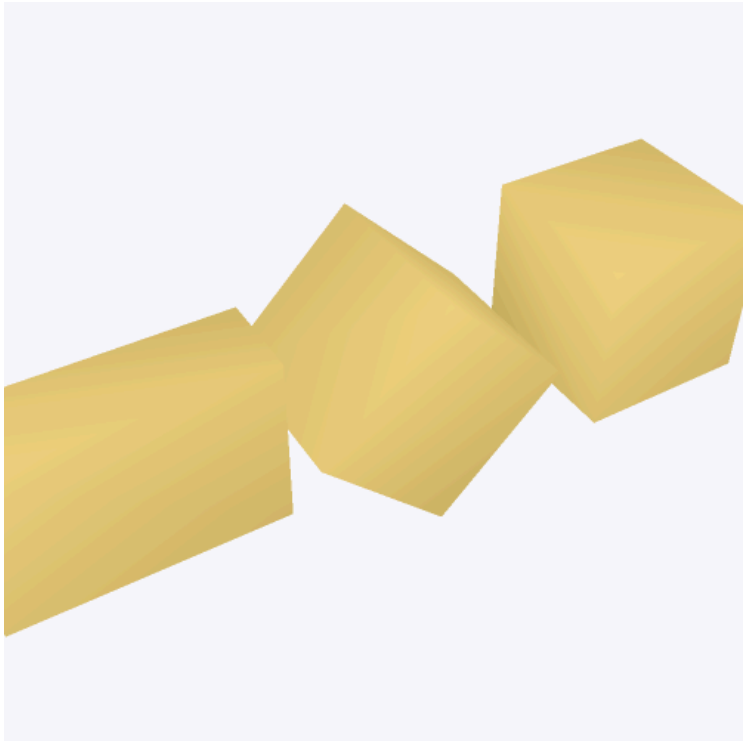
`hull` (convex hull of the union) and `minkowski` (Minkowski sum) are 3D-only.

```
#show-part(scadypst(
  hull(
    translate((-15, 0, 0), sphere(6, fn: 24)),
    translate(( 15, 0, 0), sphere(6, fn: 24)),
  )
))
```

Transforms

`translate` / `rotate` / `scale` / `mirror` / `resize` all take a child at the end.

```
#show-part(scadypst(
  union(
    cube(10, center: true),
    translate((15, 0, 0), rotate((0, 45, 0), cube(10, center:
true))),
    translate((30, 0, 0), scale((1.5, 0.5, 1), cube(10, center:
true))),
  )
))
```

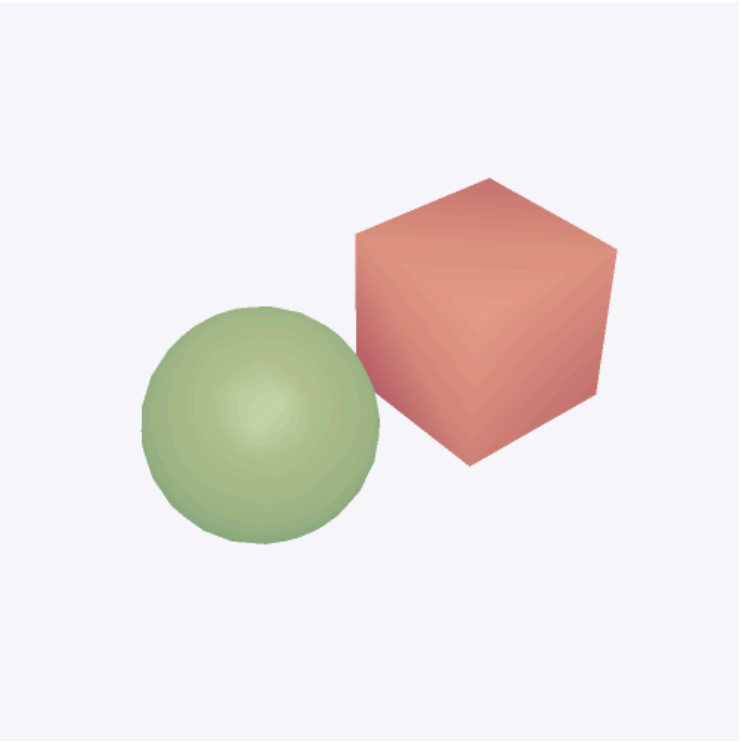


`mirror` reflects across a plane whose normal is the vector you pass.

```
#show-part(scadypst(
  union(
    translate((-8, 0, 0), cylinder(20, r1: 8, r2: 2, fn: 32)),
    mirror((1, 0, 0), translate((-8, 0, 0), cylinder(20, r1: 8,
r2: 2, fn: 32))),
  )
))
```

`color` propagates through boolean ops; the emitted PLY carries per-face RGB, which `render-ply` picks up automatically:

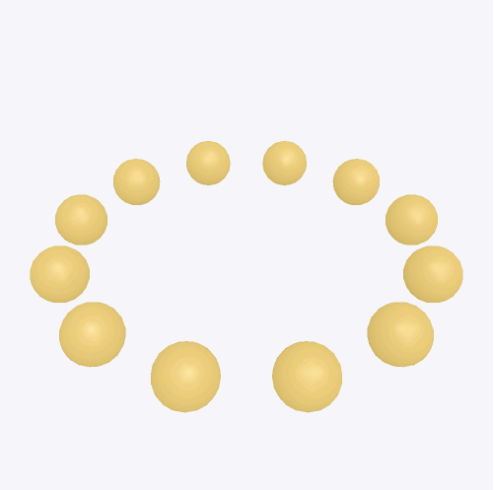
```
#show-part(
  scadypst(union(
    color((0.9, 0.3, 0.3), cube(10, center: true)),
    color((0.3, 0.7, 0.4), translate((15, 0, 0), sphere(6, fn:
32))),
  )),
  color: none,
)
```



Procedural placement (Typst `for`)

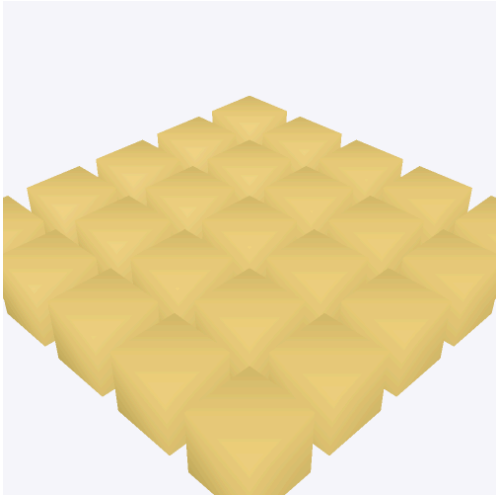
The DSL sits in Typst. You don't need OpenSCAD's `for()` — use Typst's own iteration.

```
#let ring = union(..for i in range(12) {
  let a = i * 30
  let r = 20
  (translate(
    (calc.cos(a * 1deg) * r, calc.sin(a * 1deg) * r, 0),
    sphere(3, fn: 24),
  ),)
})
#show-part(scadypst(ring))
```



Nested loops for a grid:

```
#let grid = union(..for x in range(-2, 3) {
  for y in range(-2, 3) {
    (translate(
      (x * 8, y * 8, 0),
      cube(6, center: true),
    ),)
  }
})
#show-part(scadypst(grid))
```



2D primitives

For extrusion sources. Render one by extruding it a tiny amount so we can see the outline:

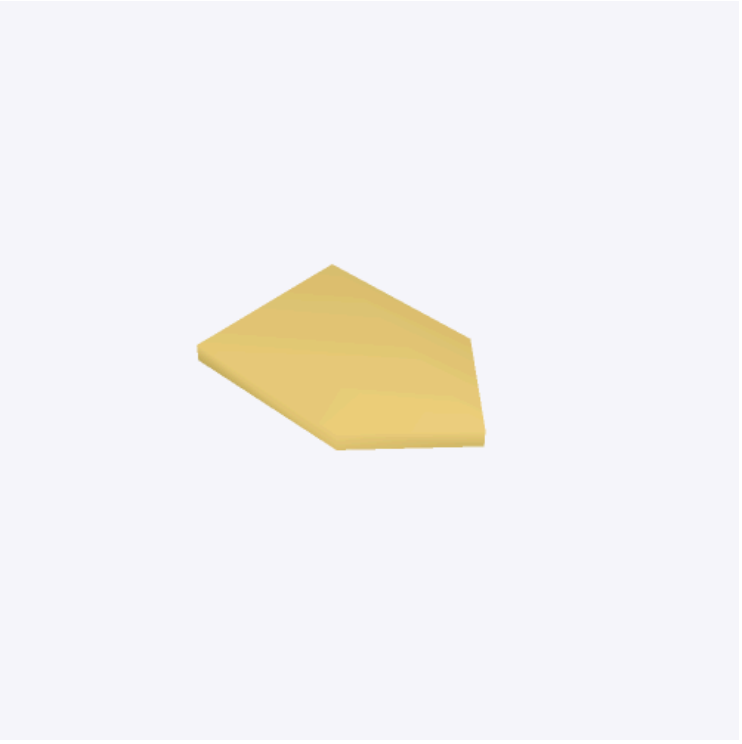
```
#show-part(scadypst(linear-extrude(1, star(6, 15, 6))))
```

```
#show-part(scadypst(linear-extrude(1, ngon(6, 10))))
```

```
#show-part(scadypst(linear-extrude(1, rounded-square(20, 12, 3,
fn: 24))))
```

`polygon` — arbitrary 2D shapes from `(x, y)` point lists. `paths` (optional) is a list of index-rings: first ring is the outer boundary, rest are holes.

```
#show-part(scadypst(linear-extrude(1,
  polygon(((0, 0), (10, 0), (10, 10), (5, 15), (0, 10))),
)))
```



Offset

`offset(d, shape2d)` grows (`d > 0`) or shrinks (`d < 0`) a 2D shape by `d` units. Useful for creating outlines / clearance.

```
#show-part(scadypst(linear-extrude(1,
  offset(2, square(10, center: true)),
)))
```

Compiling an existing `.scad` file

`compile-scad(source)` takes real OpenSCAD source (parsed via `openscad-rs`, evaluated in-crate). Full language coverage per `FIDELITY.md`.

```
#import "@preview/maquette-scad:0.1.0": compile-scad
#import "@preview/maquette:0.1.3": render-ply

#let part = compile-scad(read("part.scad"))
#render-ply(part)
```

For `.scad` files that `use` or `include` other `.scad` files, or that import an STL/OBJ mesh, supply the sidecar bytes through the `files` / `bin` / `font` params:

```
#compile-scad(read("main.scad"),
  files: (
    "utils.scad": read("utils.scad"),
    "gears.scad": read("gears.scad"),
  ),
  bin: (
    "mount.stl": read("mount.stl", encoding: none),
  ),
  font: read("logo.ttf", encoding: none),
  fn: 48,
)
```

The `fn` argument sets a default `$fn` for the compile (overridable per-primitive via `fn:` on `cube` / `sphere` / etc.). `trace: some-path-string` dumps the evaluator trace to the given path — useful when a nested `for` or `module` isn't producing what you expected.

Language coverage

The `.scad` evaluator is complete for everything a typical file uses. TL;DR:

Supported (✅) — comments, numbers, bool, string, `undef`, vectors, ranges, variable assignment (last-wins scope), `let()` expression, member access, indexing, all standard operators, `$fn` / `$t` / `$preview` / `$children`, all 2D + 3D primitives, all transforms (`translate`, `rotate`, `scale`, `mirror`, `resize`, `multmatrix`, `color`, `offset`), all booleans (`union`, `difference`, `intersection`), extrusions (`linear_extrude` with twist / scale / slices, `rotate_extrude`), `hull`, `minkowski`, `projection`, list comprehensions, `for`, `if`, `intersection_for`, `function`, `module`, `render`, string ops, math functions, `include` + `use`, `color()` with per-face RGB propagation.

Partial (🟡) — `$fa` / `$fs` are defined so libraries reading them work, but our primitive builder tessellates from `$fn` (or `fn:` on the primitive), not adaptively.

Not supported (❌) — `import()` of external STL/OBJ/DXF (wasm sandbox has no filesystem — pass bytes via `bin:` instead), `surface()` heightmap import, `$vpr` / `$vpt` / `$vpd` / `$vpf` viewport variables (n/a — no viewport at eval time), font-dependent text glyphs beyond basic Latin-1 skeletons.

See `crates/maquette-scad/FIDELITY.md` for the exhaustive per-feature tracker.

Design notes

Why Manifold? — the previous kernel we used (`csgrs` 0.20.1, BSP-based) left about 20% of faces open on any non-trivial boolean (bores, notches, partial overlaps). Manifold guarantees watertight, correctly-triangulated output. Every mesh this crate emits has zero open edges, verified on the full test corpus.

Why compile in-crate? — a `.scad`-then-shell-out design would need a filesystem, a working OpenSCAD binary on the host, and IPC across a process boundary. The wasm plugin has none of that. Ingesting `.scad` text ourselves keeps the whole compile deterministic under Typst's sandbox — same input, same PDF, cross-platform.

Two wasm modules. — `maquette-scad.wasm` (this crate) compiles geometry to PLY. `maquette.wasm` (the sibling plugin) renders the PLY. They only exchange the PLY blob; you can also send `scadypst(...)`'s output to `maquette-gltf` via a `.gltf` wrapper if you want PBR shading on procedural geometry.